A FURTHER RESULTS

This appendix collects the results that Secs. 3, 4, 5 and 6 refer to.

Unsound BCNF decompositions. The following proposition formalizes the observation from Ex. 3.5.

Proposition A.1 (Unsound BCNF decompositions). *There is a schema (R, Σ) and a lossless decomposition $\mathcal{D}$ of (R, Σ) into BCNF, computed by the classical decomposition algorithm, such that the key design $(\mathcal{D}, \kappa)$ that declares the minimal keys of $(S, \Sigma[S])$ on every $S \in \mathcal{D}$ is not sound.*

Inheritance. The following lemma shows that an uncritical FD stays uncritical when other FDs are left unenforced, which Thm. 3.21 and Alg. 2 rest on.

Lemma A.2 (Inheritance). *Let $\mathcal{E} \subseteq \Sigma^+$ be a closed FD set that contains every uncritical minimal FD of Σ. Then every uncritical minimal FD of Σ is an uncritical minimal FD of $\mathcal{E}$.*

Verification. The hardness of Thm. 4.5 rests on the representation of a design by its declared keys. A design is locally enforced iff, for every minimal FD $X \to A$ of $\mathcal{E}$ and every $S \in \mathcal{D}$ with $XA \subseteq S$, the set X contains some $K \in \kappa(S)$, which takes time polynomial in the size of the atomic cover of $\mathcal{E}$.

Shapes of weakened FDs. The following proposition shows that the enumeration of line 5 of Alg. 1 can be confined neither to one weakened form per unenforced FD nor to the left-hand sides of minimal FDs.

Proposition A.3 (Shapes of weakened FDs). *There is a schema (R, Σ) with $\lambda(R, \Sigma) = 1$ such that every BCNF design of (R, Σ) with loss one enforces two weakened forms of the same unenforced minimal FD. There is a schema (R, Σ) with $\lambda(R, \Sigma) = 2$ such that every BCNF design of (R, Σ) with loss two enforces a weakened form $X' \to A$ of an unenforced minimal FD where X' is not contained in the left-hand side of any minimal FD of Σ.*

The closures of the left-hand sides of critical minimal FDs do not confine the weakened FDs either. For $\Psi = \Sigma_a - \Delta$, call a weakened FD $X' \to A$ *local* iff $X' \subseteq Y^+_\Psi$ for some minimal FD $Y \to B$ of Ψ^+ that is critical for Ψ^+. Only the critical minimal FDs of Ψ^+ obstruct the enforceability of Ψ^+, and local weakened FDs suffice in the two schemata of Prop. A.3, but the following proposition shows that they do not suffice in general.

Proposition A.4 (Local weakened FDs). *There is a schema (R, Σ) with a set $\Delta \subseteq \Sigma_a$ and weights w such that $|\Delta| = \lambda(R, \Sigma)$, every BCNF design of (R, Σ) with loss $\lambda_w(R, \Sigma)$ leaves exactly Δ unenforced, and every BCNF design of (R, Σ) that leaves exactly Δ unenforced enforces a weakened FD that is not local.*

Hence, line 5 of Alg. 1 cannot be confined to local weakened FDs. The search of Lem. 5.6 needs no such restriction, since it measures closures against the current set $\mathcal{E}$ rather than Ψ. Under unit weights, the schema of the proof has loss three as well, but a plain design attains it by leaving $BDE \to A$, $DF \to C$ and $AF \to C$ unenforced, so the proposition needs the weights. Under unit weights, every random instance we examined has a design with minimum loss whose enforced FDs are generated by Ψ and local weakened FDs, but whether such a design always exists is open.

Tightness of the cover condition. The cover condition of Lem. 5.3 cannot be strengthened to confine Δ to the critical minimal FDs and the FDs that violate their schemata.

Proposition A.5 (Tightness of the cover condition). *There is a schema (R, Σ) such that every BCNF design of (R, Σ) with minimum loss leaves a minimal FD of Σ unenforced that is uncritical and violates the schema of no critical minimal FD of Σ.*

The optimal designs of this schema are not plain, since every plain design leaves at least four minimal FDs unenforced, while the design of the proof leaves three.

Isolated critical schemata. The following proposition identifies a case in which Alg. 3 is optimal.

Proposition A.6 (Isolated critical schemata). *If the critical schemata of $\mathcal{D}_0$ share no attribute with each other or with any other schema of $\mathcal{D}_0$, except for a schema that is a minimal key of (R, Σ), then line 9 of Alg. 3 succeeds and the returned design has loss $\lambda_w(R, \Sigma)$.*

In this case, (R, Σ) is the attribute-disjoint union of the critical schemata and of the remaining attributes, on which the other schemata form a design with loss zero, so the local optima add up to the global optimum by Lem. 4.2, and a key schema only joins the parts.

Branches of Repair. Tab. 7 lists for every FD set of Sec. 6 with positive loss the loss of every design that Alg. 3 examines before it returns the lightest. When the assembled set Γ^+ is enforceable, these are the design of Γ, the obstruction design O of Thm. 3.21 and the ascents of Alg. 2 from both. Otherwise, the fallback examines the obstruction design of (R, Γ^+) in place of the design of Γ, a greedy enforceable subset of Γ^+, the global Drop with the remaining budget, which includes its own ascents, and the ascent from the obstruction design of (R, Γ^+). A dash marks a step that its budget cut or that Drop had already run. The assembled design alone is never the lightest: on abalone and echo the ascent from it recovers the minimal FDs that no schema of iConf embeds, on the six sets of the fallback the time-limited Drop or the ascent from the obstruction design of (R, Γ^+) wins, and on breast the obstruction design of (R, Σ) is optimal as well.

B PROOFS

Proof of Lemma 3.3. Since $\mathcal{E} \subseteq \Sigma^+$, the FD $X \to A$ is in Σ^+ and $A \notin X$. Hence, there is a minimal $Y \subseteq X$ with $Y \to A \in \Sigma^+$, which is a minimal FD of Σ. □

Proof of Lemma 3.8. Dependency preservation holds since $\mathcal{E}$ is the closure of key dependencies, each of which is embedded in some schema of $\mathcal{D}$. If $(\mathcal{D}, \kappa)$ is locally enforced, then every non-trivial FD $X \to Y \in \mathcal{E}[S]$ has a left-hand side that contains some $K \in \kappa(S)$, so X is a key of S for $\mathcal{E}[S]$. For the converse, let $X \to Y \in \mathcal{E}[S]$ be non-trivial. Since $(S, \mathcal{E}[S])$ is in BCNF, X is a key of S for $\mathcal{E}[S]$, so X contains a minimal key of $(S, \mathcal{E}[S])$, which belongs to $\kappa(S)$ by assumption. □

Proof of Proposition 3.9. For statement 1, local enforcement gives $\mathcal{E}[S] = \{K \to S \mid K \in \kappa(S)\}^+[S]$, and r_S satisfies these key dependencies, so it satisfies $\mathcal{E}[S]$. By Lem. 3.8, $(S, \mathcal{E}[S])$ is in BCNF, and no relation over a BCNF schema that satisfies its FDs contains a redundant data value [59]. For statement 2, let t_1 and t_2 be tuples

Table 6: Notation

Symbol	Meaning	Symbol	Meaning
Σ, Σ^+, Σ_a	FD set, its closure, its atomic cover of minimal FDs	X^+_Σ	attribute closure of X under Σ
$(\mathcal{D}, \kappa)$	key design with decomposition $\mathcal{D}$ and declared keys $\kappa(S)$ for every $S \in \mathcal{D}$ (Def. 3.1)	$\mathcal{E}$	enforced FDs of a design, the closure of its key dependencies
λ_w, λ	dependency loss with weights w and with unit weights (Def. 3.10)	$\nu(R, \Sigma)$	number of non-key minimal FDs of Σ (Prop. 3.12)
critical	a schema of a 3NF decomposition that is not in BCNF (Sec. 2), or a minimal FD $X \to A$ of $\mathcal{E}$ such that $(XA, \mathcal{E}[XA])$ is not in BCNF (Def. 3.16)	$\mathcal{U}$, O	uncritical minimal FDs of Σ, obstructions $\Sigma_a - \mathcal{U}^+$ (Def. 3.20)
Ψ	uncritical set, which generates an enforceable FD set (Cor. 3.18)	Δ, W	minimal FDs of Σ left unenforced, weakened FDs enforced instead (Alg. 1)
$C(\Delta)$, $Q(g, \Delta)$	critical minimal FDs whose schemata Δ relieves, requirement set of g (Lem. 5.3, Prop. 5.4)	$\mathcal{D}_0$, Γ	3NF decomposition and key dependencies of the local designs in Repair (Alg. 3)

Table 7: Losses of the Designs that Repair Examines, Whether the Assembled Set Γ^+ Is Enforceable, and the Design It Returns (dash: not run)

Data set	Γ^+ enf.	Γ^+	O	greedy	Ascend(Γ^+)	Ascend(O)	Drop	returned
abalone	yes	22	11	–	8	8	–	Ascend(Γ^+)
breast	yes	5	3	–	3	3	–	Ascend(Γ^+)
bridges	no	30	28	30	11	11	11	Ascend(Γ^+)
echo	yes	150	29	–	11	11	–	Ascend(Γ^+)
ncvoter	no	248	88	241	65	–	48	Drop
weather	no	195	192	166	91	–	84	Drop
lineitem	no	424	145	423	71	–	69	Drop
uniprot	no	856	342	780	444	–	136	Drop
hepatitis	no	2,463	2,007	–	1,361	–	992	Drop

Table 8: Unenforced Minimal FDs of the Eight Key Designs that Span Several Schemata (the Rest of the Loss of Tabs. 2 and 3 Lies Inside One Schema)

Data set	key-only	BCNF	greedy	iConf	classical	Hybrid	Repair	Drop
abalone	0	95	92	2	13	17	1	1
breast	0	38	37	1	1	1	1	2
bridges	0	103	99	7	9	13	2	3
echo	0	463	392	52	39	81	1	1
ncvoter	0	704	668	59	51	248	19	19
weather	0	869	858	54	55	209	13	13
hepatitis	0	8,211	7,995	961	1,187	2,230	550	–
food	0	4	4	4	4	–	1	1
price paid	0	2	1	0	0	–	1	1
hospital	0	4	1	4	4	–	1	1
TPC-DS	0	180	51	0	0	–	0	0

of the join that agree on X, and let $X \to Y \in \mathcal{E}$. The closure $X^+_\mathcal{E}$ arises from X by repeatedly adding a schema $S \in \mathcal{D}$ that has a key $K \in \kappa(S)$ within the attributes reached so far. If t_1 and t_2 agree on these attributes, then $t_1[S]$ and $t_2[S]$ are tuples of r_S that agree on its key K, so t_1 and t_2 agree on S as well. By induction, they agree on $X^+_\mathcal{E} \supseteq Y$. Statement 3 is the definition of soundness. For the last statement, the join of a state that satisfies the keys satisfies every FD of $\mathcal{E}$ by statement 2. Conversely, if $\varphi \notin \mathcal{E}$, then some relation r over R satisfies $\mathcal{E}$ and violates φ, since $\mathcal{E}$ is closed. Its projections form a state that satisfies the keys, and their join contains r, so it violates φ. □

Proof of Proposition 3.12. The enforced FDs of $(\{R\}, \kappa)$ are the closure $\mathcal{E}$ of the key dependencies of Σ. For every $X \subseteq R$, $X^+_\mathcal{E} = R$ if X contains a minimal key and $X^+_\mathcal{E} = X$ otherwise, so every non-trivial FD in $\mathcal{E}$ has a key as left-hand side. Hence, the design is locally enforced, it is sound as $\mathcal{D} = \{R\}$, and $\Sigma_a - \mathcal{E}$ consists of the non-key minimal FDs. For the second statement, let $X \to A$ be a non-key minimal FD of Σ and suppose $X \to A \in \mathcal{E}'$ for the enforced FDs $\mathcal{E}'$ of $(\{R\}, \kappa')$. Local enforcement requires that X contains some $K \in \kappa'(R)$, so $X \to R \in \Sigma^+$, which contradicts that X is not a key. □

Proof of Theorem 3.14. Let $(\mathcal{D}, \kappa)$ be a BCNF design with $\lambda(\mathcal{D}, \kappa) = 0$ and enforced FDs $\mathcal{E}$. Then $\Sigma_a \subseteq \mathcal{E} \subseteq \Sigma^+$, so $\mathcal{E} = \Sigma^+$ as Σ_a is a cover of Σ. Then $\mathcal{D}$ is lossless for Σ by soundness, and dependency-preserving and in BCNF for Σ by Lem. 3.8. Conversely, let $\mathcal{D}$ be a lossless, dependency-preserving decomposition of (R, Σ) into BCNF, and let $\kappa(S)$ consist of the minimal keys of $(S, \Sigma[S])$ for every $S \in \mathcal{D}$. Since $(S, \Sigma[S])$ is in BCNF, every FD in $\Sigma[S]$ is implied by the key dependencies of S, so $\mathcal{E}(\mathcal{D}, \kappa)$ is the closure of $\bigcup_{S \in \mathcal{D}} \Sigma[S]$, which is Σ^+ by dependency preservation. Hence, $(\mathcal{D}, \kappa)$ is sound, locally enforced, and has loss zero. □

Proof of Proposition 3.15. Let $R_n = \{S, T, Z, P_1, \ldots, P_n\}$ and $\Sigma_n = \{ST \to Z, Z \to T, Z \to P_1, \ldots, Z \to P_n\}$. The minimal keys are ST and SZ, and the minimal FDs are $ST \to Z$, $ST \to P_i$, $Z \to T$ and $Z \to P_i$ for $i = 1, \ldots, n$. The non-key minimal FDs are $Z \to T$ and $Z \to P_i$, so the design $(\{R_n\}, \kappa)$ has loss $n+1$. A set that contains neither Z nor ST determines only itself, and a set that contains SZ or ST is a key, so every violating FD $X \to W$ of a schema has $Z \in X$ and $S \notin X$. A step of the classical algorithm on such an FD splits a schema S' into XW and $S' - W$, which both contain Z, and only $S' - W$ contains S, since no set without S determines S. As long as the schema that contains S also contains some attribute B of $TP_1 \cdots P_n$, the FD $Z \to B$ violates it, since Z does not determine S. Hence, every run ends with the schema SZ with key SZ and schemata that contain Z but not S, whose only minimal key is Z. Their keys imply $Z \to T$ and every $Z \to P_i$, while the closure of ST under them is ST, so they imply neither $ST \to Z$ nor any $ST \to P_i$, and the loss is $n + 1$. The design $\{STZ, ZP_1 \cdots P_n\}$ with keys ST and SZ of STZ and key Z of $ZP_1 \cdots P_n$ is sound since $STZ \cap ZP_1 \cdots P_n = Z$ determines $P_1 \cdots P_n$, it is locally enforced, and it enforces every minimal FD except $Z \to T$. Finally, $ST \to Z$ is the only minimal FD of Σ_n with right-hand side Z, and it is critical since $Z \to T$ is a non-key FD of $(STZ, \Sigma_n[STZ])$, so Σ^+_n is not enforceable by Thm. 3.17 and $\lambda(R_n, \Sigma_n) \geq 1$. □

Proof of Theorem 3.17. Statement 1 implies statement 2 by Lem. 3.8 and soundness. For the converse, let $\mathcal{D}$ be a lossless,

dependency-preserving decomposition of $(R, \mathcal{E})$ into BCNF, and let $\kappa(S)$ consist of the minimal keys of $(S, \mathcal{E}[S])$ for every $S \in \mathcal{D}$. As in the proof of Thm. 3.14, $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) with $\mathcal{E}(\mathcal{D}, \kappa) = \mathcal{E}$, it is sound since $\mathcal{D}$ is lossless for $\mathcal{E}$, and it is locally enforced by Lem. 3.8. Next, let $\mathcal{D}$ be as before, let $S \in \mathcal{D}$, let K be a minimal key of $(S, \mathcal{E}[S])$, and let $A \in S - K$. Some minimal $X \subseteq K$ satisfies $X \to A \in \mathcal{E}$. Every non-trivial FD $Y \to B \in \mathcal{E}[XA]$ belongs to $\mathcal{E}[S]$, so Y is a key of S for $\mathcal{E}$ since $(S, \mathcal{E}[S])$ is in BCNF, and therefore Y is a key of XA for $\mathcal{E}$. Hence, $X \to A$ is uncritical. The FDs obtained in this way imply every key dependency $K \to S$ of $\mathcal{D}$, and these imply $\mathcal{E}$ since $\mathcal{D}$ is dependency-preserving and in BCNF for $\mathcal{E}$. Finally, let the uncritical minimal FDs U of $\mathcal{E}$ form a cover of $\mathcal{E}$, and let K be a minimal key of $(R, \mathcal{E})$. The decomposition $\{XA \mid X \to A \in U\} \cup \{K\}$ is in BCNF for $\mathcal{E}$, since $(K, \mathcal{E}[K])$ has no non-trivial FD by the minimality of K. It is dependency-preserving for $\mathcal{E}$ since U is a cover of $\mathcal{E}$, and it is lossless for $\mathcal{E}$ since it is dependency-preserving and contains a key [12]. □

Proof of Corollary 3.18. Let $\mathcal{E} = \Psi^+$, let K be a minimal key of $(R, \mathcal{E})$, and let $\mathcal{D}$ consist of the schemata XA of the elements $X \to A$ of Ψ and of K. Every $(XA, \mathcal{E}[XA])$ is in BCNF since Ψ is uncritical, and $(K, \mathcal{E}[K])$ has no non-trivial FD by the minimality of K. Hence, $\mathcal{D}$ is in BCNF for $\mathcal{E}$, it is dependency-preserving for $\mathcal{E}$ since Ψ is a cover of $\mathcal{E}$ whose elements are embedded in its schemata, and it is lossless for $\mathcal{E}$ since it contains a key [12]. As in the proof of Thm. 3.17, declaring the minimal keys of $(S, \mathcal{E}[S])$ on every $S \in \mathcal{D}$ yields a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$, so $\mathcal{E}$ is enforceable. □

Proof of Lemma A.2. Let $X \to A$ be an uncritical minimal FD of Σ, and let $\mathcal{U} \subseteq \mathcal{E} \subseteq \Sigma^+$ with $\mathcal{E}$ closed. Since $\mathcal{E} \subseteq \Sigma^+$ contains $X \to A$, the FD $X \to A$ is a minimal FD of $\mathcal{E}$. Let $Y \to B \in \mathcal{E}[XA]$ be non-trivial, and let $Z \in XA - Y$. Since $Y \to B \in \Sigma^+[XA]$ and $X \to A$ is uncritical for Σ^+, Y is a key of XA for Σ, so some minimal FD $Y' \to Z$ of Σ satisfies $Y' \subseteq Y$. If $Y' \to Z$ were critical for Σ^+, some non-trivial FD $V \to W \in \Sigma^+[Y'Z]$ would have a left-hand side V that is not a key of $Y'Z$ for Σ, but $V \to W \in \Sigma^+[XA]$ since $Y'Z \subseteq XA$, so V is a key of XA for Σ and hence of $Y'Z$, a contradiction. Consequently, $Y' \to Z \in \mathcal{U} \subseteq \mathcal{E}$ and $Y \to Z \in \mathcal{E}$ for every $Z \in XA - Y$, so Y is a key of XA for $\mathcal{E}$. Hence, $(XA, \mathcal{E}[XA])$ is in BCNF, that is, $X \to A$ is uncritical for $\mathcal{E}$. □

Proof of Theorem 3.21. Let $\mathcal{E} = \mathcal{U}^+$. By Lem. A.2, every element of $\mathcal{U}$ is an uncritical minimal FD of $\mathcal{E}$, and $\mathcal{U}$ is a cover of $\mathcal{E}$. Hence, the uncritical minimal FDs of $\mathcal{E}$ form a cover of $\mathcal{E}$, and $\mathcal{E}$ is enforceable for (R, Σ) by Thm. 3.17. A design with enforced FDs $\mathcal{E}$ has loss $w(\Sigma_a - \mathcal{E}) = w(\mathcal{O})$, so $\lambda_w(R, \Sigma) \leq w(\mathcal{O})$. □

Proof of Proposition A.1. Ex. 3.5 provides the schema and the decomposition $\mathcal{D}_3$. □

Proof of Proposition 3.22. Since $\Sigma[S] \subseteq \mathcal{E} \subseteq \Sigma^+$, we have $\mathcal{E}[S] = \Sigma[S]$ for every $S \in \mathcal{D}$, so $(S, \mathcal{E}[S])$ is in BCNF and $\mathcal{E}[S]$ is implied by the key dependencies of the minimal keys of $(S, \Sigma[S])$. The key dependency of K on K is trivial, and the projection $\mathcal{E}[K]$ has no non-trivial FD by the minimality of K. Hence, the declared key dependencies imply exactly the closure $\mathcal{E}$ of $\bigcup_{S \in \mathcal{D}} \Sigma[S]$, the design enforces $\mathcal{E}$, and $\mathcal{D} \cup \{K\}$ is dependency-preserving for $\mathcal{E}$. The design is sound since $\mathcal{D} \cup \{K\}$ is dependency-preserving for $\mathcal{E}$ and contains a key [12], and it is locally enforced since $\mathcal{E}[S]$ is implied by the keys declared on S and $\mathcal{E}[K]$ is trivial. In particular, $\mathcal{E}$ is enforceable. For the general statement, the key dependency of K' on K' is trivial, so the enforced FDs remain $\mathcal{E}'$, and every schema of $\mathcal{D}'$ keeps its keys and remains locally enforced. The projection $\mathcal{E}'[K']$ contains no non-trivial FD, since an FD $Y \to B \in \mathcal{E}'$ with $YB \subseteq K'$ and $B \notin Y$ would make $K' - B$ a key of $(R, \mathcal{E}')$, so the design is locally enforced on K' as well. It is sound since $\mathcal{D}' \cup \{K'\}$ is dependency-preserving for $\mathcal{E}'$, whose generating key dependencies are embedded in its schemata, and contains the key K' of $(R, \mathcal{E}')$ [12]. □

Proof of Theorem 4.1. For $k = 0$ the statement is Thm. 3.14 together with the NP-hardness of deciding the existence of a lossless, dependency-preserving decomposition into BCNF [5]. This NP-hardness persists under the assumption of Sec. 2, since deleting the set C of constant attributes from R and from the FDs of Σ takes polynomial time and preserves the existence of such a decomposition. Indeed, a schema that contains an attribute of C and an attribute outside C violates BCNF by $\emptyset \to C$, so the schemata disjoint from C form such a decomposition of $R - C$, and conversely every such decomposition of $R - C$ extends by the schema C. For $k > 0$ we reduce from the case $k = 0$. Let (R, Σ) be an instance, and let $(R_1, \Sigma_1), \ldots, (R_k, \Sigma_k)$ be k copies of the schema $(\{S, T, Z\}, \{ST \to Z, Z \to T\})$ over pairwise disjoint sets of fresh attributes. Each copy has loss one, since its key-only design leaves only $Z \to T$ unenforced (Prop. 3.12), while its closure is not enforceable, as $ST \to Z$ is critical and the only minimal FD with right-hand side Z, and the copies share no attribute with R and introduce no constant attribute, so the union satisfies the assumption of Sec. 2. By Lem. 4.2, applied k times, $\lambda(R \cup R_1 \cup \cdots \cup R_k, \Sigma \cup \Sigma_1 \cup \cdots \cup \Sigma_k) = \lambda(R, \Sigma) + k$, so (R, Σ) is in 0-loss BCNF iff the union is in k-loss BCNF. □

Proof of Lemma 4.2. Let $R = R_1 \cup R_2$ and $\Sigma = \Sigma_1 \cup \Sigma_2$. For $X \subseteq R$, the attribute closure X^+_Σ is the union of $(X \cap R_1)^+_{\Sigma_1}$ and $(X \cap R_2)^+_{\Sigma_2}$. Hence, the minimal FDs of Σ are those of Σ_1 together with those of Σ_2, and the closure of a set $X \subseteq R_i$ under any subset of Σ^+ is contained in R_i, since an FD $Y \to B \in \Sigma^+$ with $Y \subseteq R_i$ has $B \in Y^+_\Sigma = Y^+_{\Sigma_i} \subseteq R_i$, as $\emptyset^+_\Sigma = \emptyset$ by the assumption of Sec. 2. We use the characterization of enforceable FD sets in Thm. 3.17. Let $\mathcal{E}$ be enforceable for (R, Σ), and let $\mathcal{E}_i = \mathcal{E}[R_i]$ for $i = 1, 2$. Then $\mathcal{E}_i$ is a closed subset of Σ_i^+, its minimal FDs are the minimal FDs of $\mathcal{E}$ over R_i, and such an FD $X \to A$ is critical for $\mathcal{E}_i$ iff it is critical for $\mathcal{E}$, since both properties depend on $\mathcal{E}[XA] = \mathcal{E}_i[XA]$ only. Every FD $X \to A \in \mathcal{E}_i$ is implied by the uncritical minimal FDs of $\mathcal{E}$, and its derivation uses FDs over R_i only, since the closure of X stays within R_i. Hence, $\mathcal{E}_i$ is enforceable for (R_i, Σ_i), and $\lambda(R_1, \Sigma_1) + \lambda(R_2, \Sigma_2) \leq |(\Sigma_1)_a - \mathcal{E}_1| + |(\Sigma_2)_a - \mathcal{E}_2| = |\Sigma_a - \mathcal{E}|$. Conversely, let $\mathcal{E}_i$ be enforceable for (R_i, Σ_i) with uncritical minimal FDs U_i, and let $\mathcal{E} = (\mathcal{E}_1 \cup \mathcal{E}_2)^+$. The minimal FDs of $\mathcal{E}$ are those of $\mathcal{E}_1$ together with those of $\mathcal{E}_2$, each of them is critical for $\mathcal{E}$ iff it is critical for its $\mathcal{E}_i$, and $U_1 \cup U_2$ is a cover of $\mathcal{E}$. Hence, $\mathcal{E}$ is enforceable for (R, Σ) with $|\Sigma_a - \mathcal{E}| = |(\Sigma_1)_a - \mathcal{E}_1| + |(\Sigma_2)_a - \mathcal{E}_2|$. □

Proof of Corollary 4.4. Let $\mathcal{A}$ be a polynomial-time algorithm whose designs have loss at most $\rho \cdot \lambda(R, \Sigma)$. The design returned for (R, Σ) has loss zero iff (R, Σ) is in 0-loss BCNF, and it has loss zero iff its declared key dependencies imply every FD

of Σ, which attribute closures decide in polynomial time. Hence, $\mathcal{A}$ decides 0-loss BCNF, which is NP-hard by Thm. 4.1. Now let $\mathcal{A}$ be a polynomial-time algorithm whose designs have loss at most $\lambda(R,\Sigma)+c$. Given (R,Σ), let (R',Σ') consist of $c+1$ copies of (R,Σ) over pairwise disjoint sets of attributes, and let $\mathcal{E}$ be the enforced FDs of the design that $\mathcal{A}$ returns for (R',Σ'). If (R,Σ) is in 0-loss BCNF, then $\lambda(R',\Sigma') = 0$ by Lem. 4.2, so $\mathcal{E}$ misses at most c of the minimal FDs of Σ', which are the minimal FDs of the copies, and the declared key dependencies imply every FD of some copy. Otherwise, the restriction of $\mathcal{E}$ to every copy is enforceable for that copy by the proof of Lem. 4.2, so it misses a minimal FD of the copy, and the declared key dependencies imply the FD set of no copy. Attribute closures decide which case applies in polynomial time, so $\mathcal{A}$ decides 0-loss BCNF. □

Proof of Theorem 4.3. We reduce the maximum independent set problem, which is NP-complete [20], to the decision whether $\lambda(R,\Sigma) \leq k$. Let $G=(V,E)$ be a graph with n vertices and m edges, and let α be the size of a maximum independent set of G. Let R consist of attributes P_v and Q_v for every vertex v, and of attributes G_e, C_e, D_e^1, D_e^2 and D_e^3 for every edge e. Let Σ consist of the *vertex FD* $a_v = P_v \to Q_v$ for every vertex v and, for every edge $e=uv$, of the *edge FD* $b_e = Q_uQ_vG_e \to C_e$, the *blockers* $s_e^u = P_uG_e \to C_e$ and $s_e^v = P_vG_e \to C_e$, and the *victims* $h_e^i = C_eP_uP_vD_e^i \to G_e$ for $i=1,2,3$. The schema (R,Σ) has $2n+5m$ attributes and $n+6m$ FDs, so it is computed in polynomial time. We show that $\Sigma = \Sigma_a$ and $\lambda(R,\Sigma) = 2m+n-\alpha$, so that G has an independent set of size α' iff $\lambda(R,\Sigma) \leq 2m+n-\alpha'$.

Atomic cover. The attributes P_v and D_e^i occur on no right-hand side of Σ, and Q_v occurs on the right-hand side of a_v only, so a_v is the only minimal FD with right-hand side Q_v. The attributes C_e and G_e occur only in the six FDs of the edge e, each of which has one of them on its left-hand side and the other on its right-hand side, so a set $X \subseteq R$ determines neither unless it contains one of them. Hence, X determines G_e iff $G_e \in X$ or X contains C_e, P_u, P_v and some D_e^i, and X determines C_e iff $C_e \in X$ or X contains G_e and P_u, or G_e and P_v, or G_e and one of P_u and Q_u and one of P_v and Q_v. Consequently, the victims are the minimal FDs with right-hand side G_e, the edge FD and the blockers are those with right-hand side C_e, and $\Sigma = \Sigma_a$. In particular, $P_uP_vG_e \to C_e$ is implied by Σ but not minimal, so it does not count towards the loss.

Lower bound. Let $\mathcal{E}$ be enforceable for (R,Σ), let Ψ be the set of its uncritical minimal FDs, which is a cover of $\mathcal{E}$ by Thm. 3.17, and let S be the set of vertices v with $a_v \in \mathcal{E}$. We first show that no victim h_e^i belongs to $\mathcal{E}$ if $\mathcal{E}$ contains one of s_e^u, s_e^v and $P_uP_vG_e \to C_e$. Suppose that $h_e^i \in \mathcal{E} = \Psi^+$. Then Ψ contains an FD $Z \to G_e$ with $Z \subseteq (C_eP_uP_vD_e^i)^+_{\mathcal{E}}$, and $Z \to G_e \in \Sigma^+$ shows $Z \supseteq C_eP_uP_vD_e^j$ for some j, where $j=i$ since $(C_eP_uP_vD_e^i)^+_\Sigma = C_eP_uQ_uP_vQ_vD_e^iG_e$. Each of the three FDs has its attributes in ZG_e and a left-hand side whose closure under Σ, and hence under $\mathcal{E}$, does not contain D_e^i. Hence, if one of them belongs to $\mathcal{E}$, then $(ZG_e, \mathcal{E}[ZG_e])$ is not in BCNF, which contradicts the uncriticality of $Z \to G_e$. Now consider an edge $e=uv$ and its six FDs. If $s_e^u \in \mathcal{E}$ or $s_e^v \in \mathcal{E}$, then the three victims are not in $\mathcal{E}$. Otherwise, s_e^u and s_e^v are not in $\mathcal{E}$, and if $u,v \in S$, then either $b_e \notin \mathcal{E}$ or $P_uP_vG_e \to C_e \in \mathcal{E}$, which again excludes the victims. In every case, $\mathcal{E}$ misses at least two of the six FDs, and at least three if $u,v \in S$. Together with the $n-|S|$ vertex FDs outside $\mathcal{E}$, this gives $|\Sigma_a - \mathcal{E}| \geq n-|S|+2m+e(S)$ where $e(S)$ is the number of edges with both vertices in S. Removing one vertex of every such edge from S leaves an independent set of size at least $|S|-e(S)$, so $|\Sigma_a - \mathcal{E}| \geq 2m+n-\alpha$.

Upper bound. Let I be an independent set of G, let Δ consist of the blockers of all edges and of the vertex FDs a_v with $v \notin I$, and let $\mathcal{E} = (\Sigma-\Delta)^+$. Then $\mathcal{E}$ contains no element of Δ, since Q_v occurs on no right-hand side of $\Sigma-\Delta$ for $v \notin I$, and $(P_uG_e)^+_{\mathcal{E}} \subseteq P_uQ_uG_e$ because b_e requires Q_v as well. Every element of $\Sigma-\Delta$ is uncritical for $\mathcal{E}$. This is clear for the vertex FDs, whose schemata have two attributes, and for the edge FD b_e, since the only non-trivial FDs of $\mathcal{E}$ on $Q_uQ_vG_eC_e$ have a left-hand side that contains $Q_uQ_vG_e$. For a victim h_e^i, the non-trivial FDs of $\mathcal{E}$ on its schema $C_eP_uP_vD_e^iG_e$ have a left-hand side that contains $C_eP_uP_vD_e^i$: the attributes P_u, P_v and D_e^i occur on no right-hand side of $\Sigma-\Delta$, so G_e is determined only by sets that contain $C_eP_uP_vD_e^i$, and C_e only through b_e, which requires Q_u and Q_v and thus a_u and a_v in $\mathcal{E}$, while I contains at most one of u and v. By Thm. 3.17, $\mathcal{E}$ is enforceable, and its loss is $|\Delta| = 2m+n-|I|$. Hence, $\lambda(R,\Sigma) = 2m+n-\alpha$. □

Proof of Theorem 4.5. Soundness is losslessness of $\mathcal{D}$ for the cover $\{K \to S \mid S \in \mathcal{D}, K \in \kappa(S)\}$ of $\mathcal{E}$, which the chase decides in polynomial time [1]. A witness against local enforcement is a schema $S \in \mathcal{D}$ and an FD $X \to A$ with $XA \subseteq S$ such that $X \to A \in \mathcal{E}$, which the attribute closure verifies in linear time, and $K \not\subseteq X$ for every $K \in \kappa(S)$, so local enforcement is in coNP. For the hardness, let φ be a propositional formula in conjunctive normal form with variables $x_1,\dots,x_n$ and clauses $C_1,\dots,C_m$, and let R consist of attributes P_i and Q_i for the literals x_i and $\neg x_i$ of every variable, an attribute C_j for every clause, and an attribute Z. Let $\mathcal{D}$ consist of the schema $S_0 = P_1Q_1\cdots P_nQ_nZ$ with $\kappa(S_0) = \{P_iQ_i \mid 1 \leq i \leq n\}$, of a schema LC_j with $\kappa(LC_j) = \{L\}$ for every clause C_j and every attribute L of a literal in C_j, and of the schema $S_1 = C_1\cdots C_mZ$ with $\kappa(S_1) = \{C_1\cdots C_m\}$, and let Σ be the set of these key dependencies. Then $(\mathcal{D},\kappa)$ is a key design of (R,Σ) with $\mathcal{E} = \Sigma^+$, its size is polynomial in the size of φ, and it is sound since S_0 contains the key P_1Q_1 of (R,Σ). No attribute C_j or Z occurs in a left-hand side of Σ other than $C_1\cdots C_m$, so the only non-trivial FDs of $\mathcal{E}$ on a schema LC_j or on S_1 are implied by its declared key, and these schemata are locally enforced. Let $X \subseteq S_0$ contain no declared key of S_0, that is, at most one of P_i and Q_i for every i, so that the literals of X are consistent. The closure of X under Σ consists of X, of the attributes C_j of the clauses that contain a literal of X, and of Z iff every clause contains a literal of X. Hence, $\mathcal{E}$ contains a non-trivial FD on S_0 with left-hand side X iff $Z \notin X$ and the literals of X satisfy φ. Consequently, $(\mathcal{D},\kappa)$ is locally enforced iff φ is unsatisfiable. □

Proof of Theorem 5.2. For the first statement, let $R = \{A,B,C,D,E\}$ and $\Sigma = \{D \to C, BE \to C, BD \to A, ABC \to D\}$. The minimal keys of (R,Σ) are ABE and BDE, and Σ_a consists of $BD \to A$, $D \to C$, $BE \to C$, $ABC \to D$ and $ABE \to D$. The design $\mathcal{D} = \{ABCD, BCE, BDE\}$ with $\kappa(ABCD) = \{ABC, BD\}$, $\kappa(BCE) = \{BE\}$ and $\kappa(BDE) = \{BDE\}$ enforces $\mathcal{E} = \{ABC \to D, BD \to AC, BE \to C\}^+$. It is sound since BDE is a key of $(R,\mathcal{E})$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key as left-hand side, and its loss is one since $\mathcal{E}$ contains

$ABE \to D$ but not $D \to C$. The loss is minimum because Σ^+ is not enforceable: $ABC \to D$ and $ABE \to D$ are critical for Σ^+, since $D \to C$ is a non-key FD of $(ABCD, \Sigma[ABCD])$ and $BD \to A$ is a non-key FD of $(ABDE, \Sigma[ABDE])$, and the uncritical minimal FDs $BD \to A$, $D \to C$ and $BE \to C$ do not imply $ABC \to D$. A plain design with loss one enforces $\Psi = \Sigma_a - \{\varphi\}$ for some minimal FD φ with $\varphi \notin \Psi^+$ and Ψ^+ enforceable. For $\varphi = ABE \to D$, the FDs $BE \to C$ and $ABC \to D$ imply φ. For $\varphi = D \to C$, the FD $ABC \to D$ is critical for Ψ^+ since $BD \to A$ is a non-key FD of $(ABCD, \Psi^+[ABCD])$, and the uncritical minimal FDs $BD \to A$ and $BE \to C$ of Ψ^+ do not imply it. For $\varphi = ABC \to D$, the FD $ABE \to D$ is critical for Ψ^+ since $BD \to A$ is a non-key FD of $(ABDE, \Psi^+[ABDE])$, and no other minimal FD of Ψ^+ has D as right-hand side. For $\varphi = BE \to C$, both $ABC \to D$ and $ABE \to D$ are critical for Ψ^+ by the same two non-key FDs, and no other minimal FD of Ψ^+ has D as right-hand side. For $\varphi = BD \to A$, the FD $ABC \to D$ is critical for Ψ^+ since $D \to C$ is a non-key FD of $(ABCD, \Psi^+[ABCD])$, and the uncritical minimal FDs $ABE \to D$, $D \to C$ and $BE \to C$ of Ψ^+ do not imply it. Hence, every plain design has a loss of at least two.

For the second statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{B \to E, CE \to A, CE \to B, ADE \to B, ABD \to C\}$. The minimal keys of (R, Σ) are ABD, ADE, BCD and CDE, and Σ_a consists of $B \to E$, $BC \to A$, $CE \to A$, $CE \to B$, $ADE \to B$, $ABD \to C$ and $ADE \to C$. The design $\mathcal{D} = \{ACE, BCE, ABDE, ABCD\}$ with the single keys CE, CE, ADE and ABD enforces $\mathcal{E} = \{CE \to AB, ADE \to B, ABD \to C\}^+$, which contains $ADE \to C$ but neither $B \to E$ nor $BC \to A$. It is sound since ADE is a key of $(R, \mathcal{E})$ contained in $ABDE$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has the declared key as left-hand side, and its loss is two. The loss is minimum. Indeed, each of $BC \to A$, $CE \to A$, $ADE \to B$, $ABD \to C$ and $ADE \to C$ is implied by the other minimal FDs, so a design that leaves only one of them unenforced does not exist. A design that leaves only $B \to E$ unenforced enforces $BC \to A$ and $CE \to A$, which remain non-key FDs of every schema that contains $ABCD$ or $ACDE$, so every minimal FD with right-hand side C is critical and $ABD \to C$ cannot be enforced. A design that enforces $B \to E$, in particular one with loss zero or one that leaves only $CE \to B$ unenforced, has the non-key FD $B \to E$ on every schema that contains BE properly. Every minimal FD of its enforced FDs with right-hand side B has E on its left-hand side and is therefore critical, so by Thm. 3.17 the design enforces neither $CE \to B$ nor $ADE \to B$. Hence, $\lambda(R, \Sigma) = 2$. Now let $(\mathcal{D}', \kappa')$ be a key-complete BCNF design with enforced FDs $\mathcal{E}'$, and suppose its loss is at most two. We use that C and E determine nothing but themselves, that CE determines $ABCE$ only, and that B determines BE only. Suppose first that $B \to E \in \mathcal{E}'$. Then no schema of $\mathcal{D}'$ contains BE properly, since B is not a key of such a schema. The key dependencies of $\mathcal{D}'$ derive B from CE or from ADE only through a schema that contains BE properly, so $CE \to B$ and $ADE \to B$ are both unenforced. They derive C from ABD only through a schema that contains $ABCD$ or $ACDE$, on which $BC \to A$ or $CE \to A$ would violate local enforcement, so a third minimal FD is unenforced. Suppose next that $B \to E \notin \mathcal{E}'$ and $CE \to B \in \mathcal{E}'$. Then some schema $S^* \in \{BCE, ABCE\}$ with $CE \in \kappa'(S^*)$ belongs to $\mathcal{D}'$. Since BC is a minimal key of $(S^*, \Sigma[S^*])$, key-completeness gives $BC \in \kappa'(S^*)$ and $BC \to E \in \mathcal{E}'$. If $CE \to A \notin \mathcal{E}'$, then $BC \to A \notin \mathcal{E}'$ either, as $BC \to A$ and $CE \to B$ would imply $CE \to A$, and three minimal FDs are unenforced. If $CE \to A \in \mathcal{E}'$, then $BC \to A \in \mathcal{E}'$, and C is derived from ABD only through a schema that contains $ABCD$ or $ACDE$, which $BC \to A$ or $CE \to A$ prevent, so $ABD \to C$ and $ADE \to C$ are unenforced. Suppose finally that neither $B \to E$ nor $CE \to B$ is in $\mathcal{E}'$. Then $BC \to A$ and $CE \to A$ must be enforced, and C is again derived from ABD only through a schema that contains $ABCD$ or $ACDE$, so $ABD \to C$ is unenforced. In all cases the loss is at least three. □

Proof of Lemma 5.3. Let $\mathcal{U}_\mathcal{E}$ denote the uncritical minimal FDs of $\mathcal{E}$, which form a cover of $\mathcal{E}$ by Thm. 3.17. Since $\Sigma_a \subseteq \mathcal{E} \cup \Delta$, the set $\mathcal{U}_\mathcal{E} \cup \Delta$ is a cover of Σ, so it suffices to show that $\mathcal{U} \cup \mathcal{C}(\Delta) \cup \Delta$ implies every $Y \to B \in \mathcal{U}_\mathcal{E}$. Firstly, let $Y \to B$ be a minimal FD of Σ that is critical for Σ^+, so that some minimal FD of Σ violates YB. If some such FD $Z \to C$ belonged to $\mathcal{E}$, then $Z \to C \in \mathcal{E}[YB]$ where Z is not a key of YB for $\mathcal{E} \subseteq \Sigma^+$, so $(YB, \mathcal{E}[YB])$ would not be in BCNF, which contradicts $Y \to B \in \mathcal{U}_\mathcal{E}$. Hence, every minimal FD of Σ that violates YB belongs to Δ, that is, $Y \to B \in \mathcal{C}(\Delta)$. Secondly, let $Y \to B$ not be a minimal FD of Σ. Then $Y' \to B$ is a minimal FD of Σ for some $Y' \subset Y$, and $Y' \to B \notin \mathcal{E}$ since $Y \to B$ is minimal for $\mathcal{E}$. Hence, $Y' \to B \in \Delta$ implies $Y \to B$. Every remaining element of $\mathcal{U}_\mathcal{E}$ is a minimal FD of Σ that is uncritical for Σ^+, that is, an element of $\mathcal{U}$. For the second claim, suppose that Δ contains neither a critical minimal FD of Σ nor an FD that violates the schema of one. Then $\mathcal{C}(\Delta) = \emptyset$ and $\Delta \subseteq \mathcal{U}$, so $\mathcal{U}$ is a cover of Σ and Σ^+ is enforceable by Thm. 3.17. □

Proof of Proposition 5.4. Let $(\mathcal{D}, \kappa)$ be a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$ such that $\Delta' = \Sigma_a - \mathcal{E}$ contains Δ, and let $g_i = Y \to B$. By Lem. 5.3, g_i is implied by $\mathcal{U} \cup \mathcal{C}(\Delta') \cup \Delta'$, and the attribute closure of Y under this set applies only FDs whose left-hand sides are contained in Y^+_Σ. Since $\mathcal{U} \cup \mathcal{C}(\Delta) \cup \Delta$ does not imply g_i, some FD f applied in the closure is not in this set, and $f \in \mathcal{C}(\Delta') \cup \Delta'$. If $f \in \Delta'$, then f belongs to $\mathcal{Q}(g_i, \Delta)$ by the first clause of its definition. Otherwise, $f = X \to A$ is a critical minimal FD of Σ with $X \subseteq Y^+_\Sigma$ and $f \in \mathcal{C}(\Delta') - \mathcal{C}(\Delta)$, so every minimal FD of Σ that violates XA belongs to Δ', and some such FD v is not in Δ since $f \notin \mathcal{C}(\Delta)$. Hence, v belongs to $\mathcal{Q}(g_i, \Delta)$ by the second clause, and $v \in \Delta'$. In both cases Δ' meets $\mathcal{Q}(g_i, \Delta)$, which is impossible if the requirement set is empty. Since the requirement sets are pairwise disjoint and disjoint from Δ, the set Δ' contains Δ and one distinct element of every $\mathcal{Q}(g_i, \Delta)$, which gives the bound on $w(\Delta')$. □

Proof of Theorem 5.5. A returned pair $(\mathcal{D}, \kappa)$ is a key design since every minimal key of $(S, \mathcal{E}[S])$ is a key of S for Σ. It is sound and locally enforced by Thm. 3.17 applied to $\mathcal{E}$, and its loss is $w(\Delta)$ by line 7. For optimality, let $\mathcal{E}^*$ be the enforced FDs of a BCNF design with minimum loss, and let $\Delta^* = \Sigma_a - \mathcal{E}^*$. By Lem. 3.3, every minimal FD of $\mathcal{E}^*$ either belongs to $\Sigma_a - \Delta^*$ or is a weakening of some minimal FD $X \to A$ of Σ with $X \to A \notin \mathcal{E}^*$, that is, an element of $\mathcal{W}(\Delta^*)$. Hence, $\mathcal{E}^* = (\Psi^* \cup W^*)^+$ for $\Psi^* = \Sigma_a - \Delta^*$ and the set W^* of minimal FDs of $\mathcal{E}^*$ in $\mathcal{W}(\Delta^*)$, and $\mathcal{E}^* \cap \Delta^* = \emptyset$. By Lem. 5.3, Δ^* passes the test in line 3. Since $\mathcal{E}^*$ is enforceable, the combination of Δ^* and W^* causes a return, unless the algorithm returns earlier with a loss of at most $w(\Delta^*)$. The restriction to $W = \emptyset$ examines the sets Ψ^+ with $\Psi \subseteq \Sigma_a$ and $\Psi^+ \cap \Sigma_a = \Psi$ whose complement

$\Sigma_a - \Psi$ passes line 3, and these include the enforced FDs of every plain design by Lem. 5.3. □

Proof of Lemma 5.6. Let $\mathcal{U}_\mathcal{E}$ and $\mathcal{U}_{\mathcal{E}'}$ denote the uncritical minimal FDs of $\mathcal{E}$ and $\mathcal{E}'$. By Thm. 3.17, $\mathcal{U}_{\mathcal{E}'}$ implies g, so the computation of the closure of Y under $\mathcal{U}_{\mathcal{E}'}$ derives B. Since $\mathcal{U}_\mathcal{E}$ does not imply g, this computation applies some FD of $\mathcal{U}_{\mathcal{E}'} - \mathcal{U}_\mathcal{E}$, and we let $f = V \to C$ be the first one. The FDs applied before f belong to $\mathcal{U}_\mathcal{E} \subseteq \mathcal{E}$, so $V \subseteq Y^+_\mathcal{E}$. Firstly, let $f \notin \mathcal{E}$. By Lem. 3.3, some minimal FD $X \to C$ of Σ satisfies $X \subseteq V$. It belongs to Δ, since otherwise $X \to C \in \mathcal{E}$ and hence $f \in \mathcal{E}$, and $X \neq V$, since $f \in \mathcal{E}'$ and $\mathcal{E}' \cap \Delta = \emptyset$. Hence, f is the required FD. Secondly, let $f \in \mathcal{E}$. Then f is a minimal FD of $\mathcal{E}$, since some $V' \subset V$ with $V' \to C \in \mathcal{E} \subseteq \mathcal{E}'$ would contradict the minimality of f in $\mathcal{E}'$, and f is critical for $\mathcal{E}$ since $f \notin \mathcal{U}_\mathcal{E}$. So some non-trivial FD $Z \to D \in \mathcal{E}[VC]$ has a left-hand side Z that is not a key of VC for $\mathcal{E}$, while Z is a key of VC for $\mathcal{E}'$ since f is uncritical for $\mathcal{E}'$. Hence, $Z \to a \in \mathcal{E}' - \mathcal{E}$ for some $a \in VC$, and a minimal $Z' \subseteq Z$ with $Z' \to a \in \mathcal{E}'$ yields a minimal FD $Z' \to a$ of $\mathcal{E}'$ that is not in $\mathcal{E}$. As before, Lem. 3.3 provides some $X \to a \in \Delta$ with $X \subset Z'$, and $Z' \subseteq VC \subseteq Y^+_\mathcal{E}$ since $V \subseteq Y^+_\mathcal{E}$ and $f \in \mathcal{E}$. □

Proof of Proposition A.3. For the first statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{B \to A, BC \to E, BD \to E, ACE \to B, ADE \to B\}$, whose minimal FDs are the elements of Σ. The design $\{ABCE, ABDE, BCD\}$ with $\kappa(ABCE) = \{ACE, BC\}$, $\kappa(ABDE) = \{ADE, BD\}$ and $\kappa(BCD) = \{BCD\}$ enforces $\mathcal{E} = \{ACE \to B, ADE \to B, BC \to AE, BD \to AE\}^+$. It is sound since BCD is a key of $(R, \mathcal{E})$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key as left-hand side, and its loss is one since $B \to A$ is the only minimal FD outside $\mathcal{E}$. Every BCNF design with loss one leaves $B \to A$ unenforced. Indeed, if one of the other four minimal FDs is the only unenforced one, then $B \to A$ is a non-key FD of $ABCE$ and of $ABDE$, so every FD with right-hand side B is critical and $ACE \to B$ or $ADE \to B$ cannot be enforced. With $B \to A$ unenforced, $ACE \to B$ and $ADE \to B$ are the only FDs with right-hand side B, so both must be uncritical, which requires that BC is a key of $ABCE$ and that BD is a key of $ABDE$. Hence, $BC \to A$ and $BD \to A$ are enforced, and both are minimal FDs of the enforced set since neither C nor D determines A.

For the second statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{A \to D, DE \to A, AE \to B, BCE \to D\}$, whose minimal FDs are the elements of Σ together with $DE \to B$ and $BCE \to A$. The design $\{ABCE, ABD, ADE, BDE\}$ with the single keys BCE, AB, DE and DE enforces $\mathcal{E} = \{BCE \to A, AB \to D, DE \to AB\}^+$, which contains $BCE \to D$ but neither $A \to D$ nor $AE \to B$. It is sound since BCE is a key of $(R, \mathcal{E})$, it is locally enforced, and its loss is two. An exhaustive enumeration of the closed FD sets between Σ^+ and the trivial FD set shows that every enforceable FD set with loss two leaves exactly $A \to D$ and $AE \to B$ unenforced. Such an FD set contains $BCE \to D$, which is critical since $DE \to B$ is a non-key FD of $BCDE$, so $BCE \to D$ is derived from $BCE \to A$ and some weakened form $X' \to D$ of $A \to D$ with $A \in X' \subseteq ABCE$. If $X' \subseteq AE$, then $X' \to D$ and $DE \to B$ imply the unenforced FD $AE \to B$. Hence, X' contains B or C, and no minimal FD of Σ has a left-hand side that contains AB or AC. □

Proof of Proposition A.4. Let $R = \{A, B, C, D, E, F\}$ and $\Sigma = \{DF \to C, ACE \to D, AE \to B, AF \to D, BDE \to A, DE \to F\}$. The atomic cover Σ_a consists of the elements of Σ and the minimal FDs $DE \to C$, $AF \to C$ and $ACE \to F$. Let $\Delta = \{AE \to B, AF \to C, AF \to D\}$, let $\Psi = \Sigma_a - \Delta$, and let w assign weight one to the elements of Δ and weight four to the other six minimal FDs. A closure computation shows that the elements of Ψ are the minimal FDs of Ψ^+, that only $ACE \to D$ among them is critical for Ψ^+, since $DE \to C$ violates $ACDE$, and that $ACE^+_\Psi = ACDEF$.

We first show that an enforceable FD set $\mathcal{E} \subseteq \Sigma^+$ contains neither both $AE \to B$ and $BDE \to A$, nor all of $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$. Since $\mathcal{E} \subseteq \Sigma^+$, the left-hand side of every non-trivial FD in $\mathcal{E}$ with right-hand side A contains BDE, with right-hand side D it contains ACE or AF, and with right-hand side F it contains DE or ACE. If $AE \to B$ and $BDE \to A$ are in $\mathcal{E}$, then $BDE \to A$ is the only minimal FD of $\mathcal{E}$ with right-hand side A, and it is critical for $\mathcal{E}$ since $AE \to B$ violates $ABDE$. Hence, the uncritical minimal FDs of $\mathcal{E}$ do not imply $BDE \to A$, and $\mathcal{E}$ is not enforceable by Thm. 3.17. If $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$ are in $\mathcal{E}$, then $ACE \to D$ and $ACE \to F$ are minimal FDs of $\mathcal{E}$ that are critical for $\mathcal{E}$, since $DE \to C$ violates $ACDE$ and $AF \to C$ violates $ACEF$. Every other minimal FD of $\mathcal{E}$ with right-hand side D has F in its left-hand side, and vice versa. Hence, the uncritical minimal FDs of $\mathcal{E}$ do not imply $ACE \to D$, and $\mathcal{E}$ is not enforceable.

Next, the design $\mathcal{D} = \{ABDE, ABCE, CDE, ABCF, CDF, ABDF, ACEF, DEF\}$ with $\kappa(ABDE) = \{ADE, BDE\}$, $\kappa(ABCE) = \{ACE\}$, $\kappa(ABCF) = \kappa(ABDF) = \{ABF\}$ and the single keys DE, DF, ACE and DE of CDE, CDF, $ACEF$ and DEF enforces $\mathcal{E} = (\Psi \cup \{ACE \to B, ADE \to B, ABF \to C, ABF \to D\})^+$. It is sound since ACE is a key of $(R, \mathcal{E})$ contained in $ACEF$ and $\mathcal{D}$ is dependency-preserving for $\mathcal{E}$, and it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key in its left-hand side. It leaves exactly Δ unenforced, since AE and AF are closed for $\mathcal{E}$, so its loss is $w(\Delta) = 3$. Now let $\mathcal{E}'$ be the enforced FDs of a BCNF design of (R, Σ) with loss less than four, so that its unenforced FDs form a subset Δ' of Δ. If $AE \to B \in \mathcal{E}'$, then $\mathcal{E}'$ contains $AE \to B$ and $BDE \to A$. If $AE \to B \notin \mathcal{E}'$ and $AF \to D \in \mathcal{E}'$, then $AF \to C \in \mathcal{E}'$ as well, so $\mathcal{E}'$ contains $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$. If $AE \to B$ and $AF \to D$ are not in $\mathcal{E}'$ but $AF \to C$ is, then $\mathcal{E}'$ contains the same four FDs. In each case $\mathcal{E}'$ is not enforceable, so $\Delta' = \Delta$. Hence, $\lambda_w(R, \Sigma) = 3$, and every BCNF design with this loss leaves exactly Δ unenforced. Under unit weights, let Δ' be the set of unenforced FDs of a BCNF design with enforced FDs $\mathcal{E}'$, and suppose that $|\Delta'| \leq 2$. Then Δ' contains $AE \to B$ or $BDE \to A$, and the other element of Δ', if any, is none of $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$, since each of these is implied by the elements of $\Sigma_a - \Delta'$. Hence, $\mathcal{E}'$ contains these four FDs and is not enforceable, so $\lambda(R, \Sigma) = 3 = |\Delta|$.

Finally, let $\mathcal{E}'$ be the enforced FDs of a BCNF design of (R, Σ) that leaves exactly Δ unenforced. Then $ACE \to D$ is a minimal FD of $\mathcal{E}'$ that is critical for $\mathcal{E}'$, since $DE \to C \in \mathcal{E}'$. By Thm. 3.17, the uncritical minimal FDs of $\mathcal{E}'$ imply $ACE \to D$, so D is derived from ACE through some uncritical minimal FD $X \to D$ of $\mathcal{E}'$ with $X \neq ACE$. Then X does not contain ACE and is a proper superset of AF, since $X \to D$ is minimal and $AF \to D \notin \mathcal{E}'$. If $C \in X$, then $DF \to C$ violates XD, and if $E \in X$, then $DE \to F$ violates XD,

since neither DF nor DE determines A for Σ. Hence, $X = ABF$, and $ABF \to D$ is not local since $B \notin ACE^+_\Psi$. □

Proof of Proposition A.5. Let $R = \{A, B, C, D, E\}$ and $\Sigma = \{A \to B, A \to D, D \to A, B \to E, E \to B, BC \to A\}$. The atomic cover Σ_a consists of $A \to B$, $A \to D$, $A \to E$, $D \to A$, $D \to B$, $D \to E$, $B \to E$, $E \to B$, $BC \to A$, $BC \to D$, $CE \to A$ and $CE \to D$. The minimal FDs with two attributes are uncritical. The minimal FD $BC \to A$ is critical since $A \to B$ violates ABC, and $A \to B$ is the only minimal FD that violates ABC. Likewise, $BC \to D$, $CE \to A$ and $CE \to D$ are critical, and their schemata are violated by $D \to B$, $A \to E$ and $D \to E$ only. Let G consist of these four critical minimal FDs and the four FDs that violate their schemata. The uncritical minimal FDs $\mathcal{U} = \Sigma_a - \{BC \to A, BC \to D, CE \to A, CE \to D\}$ do not imply $BC \to A$ since $BC^+_\mathcal{U} = BCE$, so Σ^+ is not enforceable by Thm. 3.17.

Let $\mathcal{D} = \{ABD, ADE, BE, BCD, CDE\}$ with $\kappa(ABD) = \{A, BD\}$, $\kappa(ADE) = \{A, DE\}$, $\kappa(BE) = \{B, E\}$, $\kappa(BCD) = \{BC\}$ and $\kappa(CDE) = \{CE\}$, and let $\mathcal{E}$ be the closure of the declared key dependencies. Then $D^+_\mathcal{E} = D$, so $D \to A$, $D \to B$ and $D \to E$ are not in $\mathcal{E}$, while every other minimal FD of Σ is, as $BC \to A$ follows from $BC \to D$ and $BD \to A$, and $CE \to A$ follows from $CE \to D$ and $DE \to A$. The decomposition is dependency-preserving for $\mathcal{E}$ and contains the key BC of $(R, \mathcal{E})$, so it is lossless for $\mathcal{E}$ and the design is sound. It is locally enforced since $\mathcal{E}[ABD] = \{A \to BD, BD \to A\}^+$, $\mathcal{E}[ADE] = \{A \to DE, DE \to A\}^+$, $\mathcal{E}[BE] = \{B \to E, E \to B\}^+$, $\mathcal{E}[BCD] = \{BC \to D\}^+$ and $\mathcal{E}[CDE] = \{CE \to D\}^+$, as C and D determine nothing and B and E determine only each other. Hence, $\lambda(R, \Sigma) \leq 3$.

Let $\mathcal{E}'$ be enforceable for (R, Σ) with $\Delta = \Sigma_a - \mathcal{E}'$. Exchanging A with D or B with E maps Σ to a cover of itself, and the two exchanges act transitively on the critical minimal FDs and on the FDs that violate their schemata. By Lem. 5.3, Δ contains an element of G, so it suffices to consider $A \to B \in \Delta$ and $BC \to A \in \Delta$. If $A \to B \in \Delta$, then $\mathcal{E}'$ contains neither both $A \to D$ and $D \to B$ nor both $A \to E$ and $E \to B$, since $\mathcal{E}'$ is closed under implication and $A \to B \notin \mathcal{E}'$. If $BC \to A \in \Delta$, then $\mathcal{E}'$ contains neither both $BC \to D$ and $D \to A$ nor both $B \to E$ and $CE \to A$. In both cases $|\Delta| \geq 3$, so $\lambda(R, \Sigma) = 3$. Finally, let $|\Delta| = 3$ and $\Delta \subseteq G$. In the first case, $\Delta = \{A \to B, D \to B, A \to E\}$ since $A \to D$ and $E \to B$ are not in G, but $D \to E$ and $E \to B$ belong to $\mathcal{E}'$ and imply $D \to B$. In the second case, $\Delta = \{BC \to A, BC \to D, CE \to A\}$ since $D \to A$ and $B \to E$ are not in G, but $CE \to D$ and $D \to A$ belong to $\mathcal{E}'$ and imply $CE \to A$. Both cases contradict $\Delta = \Sigma_a - \mathcal{E}'$, so every design with loss three leaves an element of $\Sigma_a - G$ unenforced, and every element of $\Sigma_a - G$ is uncritical and violates the schema of no critical minimal FD. □

Proof of Proposition 5.8. The uncritical minimal FDs of the enforceable set $\mathcal{E}$ form a cover of $\mathcal{E}$ by Thm. 3.17, and they form an uncritical set, since each of them is uncritical for $\mathcal{E}$. Every set Ψ that the algorithm maintains is therefore uncritical: the initial set by the previous sentence, every set of a climb by the tests in lines 5 and 9, and every kick by line 12. Line 5 adds an FD $g \in \Sigma_a - \Psi^+$, which enlarges Ψ^+ and enforces g, so $w(\Sigma_a - \Psi^+)$ decreases by at least $w(g)$, and line 9 changes Ψ only if $w(\Sigma_a - \Psi^+)$ does not increase, and at most s times in a row without decreasing it. Since the subsets of Σ^+ are finite in number, every climb terminates. Line 12 replaces Ψ^* only by a set of at most its loss, so $w(\Sigma_a - \Psi^{*+})$ never increases from $w(\Sigma_a - \mathcal{E})$, the loss of the initial set, and it decreases only finitely often, after which k kicks end the algorithm. Finally, Cor. 3.18 turns the uncritical set Ψ^* into a BCNF design of (R, Σ) with enforced FDs Ψ^{*+}, whose loss is $w(\Sigma_a - \Psi^{*+})$ by definition, at any time. □

Proof of Proposition A.6. Let $S_1, \ldots, S_m$ be the critical schemata of $\mathcal{D}_0$ and let $T = R - (S_1 \cup \cdots \cup S_m)$, so that $S_1, \ldots, S_m, T$ partition R and every schema of $\mathcal{D}_0$ other than a key schema lies inside one part. The projection of Σ onto a minimal key K of (R, Σ) is trivial, since an FD $Y \to B$ with $Y \subset K$ and $B \in K - Y$ would make $K - B$ a key. Since $\mathcal{D}_0$ is dependency-preserving for Σ, the set Σ^+ is therefore the closure of $\Sigma[S_1] \cup \cdots \cup \Sigma[S_m] \cup \Sigma[T]$, and the attribute closure of a set inside a part stays inside the part. By Lem. 4.2, applied m times, $\lambda_w(R, \Sigma) = \sum_{i=1}^m \lambda_w(S_i, \Sigma[S_i]) + \lambda_w(T, \Sigma[T])$, where the weighted version of the lemma holds with the same proof. The schemata of $\mathcal{D}_0$ inside T are in BCNF and their projections generate $\Sigma[T]$, and $K \cap T$ is a minimal key of $(T, \Sigma[T])$ since closures stay inside the parts, so these schemata and $K \cap T$ form a lossless, dependency-preserving decomposition of $(T, \Sigma[T])$ into BCNF, $\lambda_w(T, \Sigma[T]) = 0$ by Thm. 3.14, and the minimal keys of the schemata inside T generate $\Sigma[T]$. Alg. 3 adds to Γ these key dependencies, the trivial key dependency of a key schema if $\mathcal{D}_0$ has one, and, for every S_i, the key dependencies of a design of $(S_i, \Sigma[S_i])$ with loss $\lambda_w(S_i, \Sigma[S_i])$ by Thm. 5.5, whose enforced FDs $\mathcal{E}_i$ are enforceable for $(S_i, \Sigma[S_i])$. As in the proof of Lem. 4.2, the closure Γ^+ of the union of the enforceable sets $\mathcal{E}_1, \ldots, \mathcal{E}_m, \Sigma[T]$ over attribute-disjoint schemata is enforceable for (R, Σ), so line 9 succeeds by Thm. 3.17, and $w(\Sigma_a - \Gamma^+) = \sum_i w((\Sigma[S_i])_a - \mathcal{E}_i) = \lambda_w(R, \Sigma)$. The design of line 11 has loss at most $w(\Sigma_a - \Gamma^+)$ by Prop. 5.8 and at least $\lambda_w(R, \Sigma)$ by definition, which proves the claim. □

Proof of Theorem 5.10. Every element of Γ is a key dependency of a key design of some $(S, \Sigma[S])$, so $\Gamma \subseteq \Sigma^+$. If line 9 returns a lossless, dependency-preserving decomposition $\mathcal{D}$ of (R, Γ^+) into BCNF, then $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) since every minimal key of $(S, \Gamma^+[S])$ is a key of S for Σ, its enforced FDs are Γ^+ since $\mathcal{D}$ is dependency-preserving and in BCNF for Γ^+, and it is sound and locally enforced as in the proof of Thm. 5.5. Its loss is $w(\Sigma_a - \Gamma^+)$ by definition, and the obstruction design is a BCNF design of (R, Σ) with loss $w(\mathcal{O})$ by Thm. 3.21, so line 11 returns a BCNF design whose loss is at most the minimum of the two by Prop. 5.8. Otherwise, the claim follows from Thm. 5.5. □